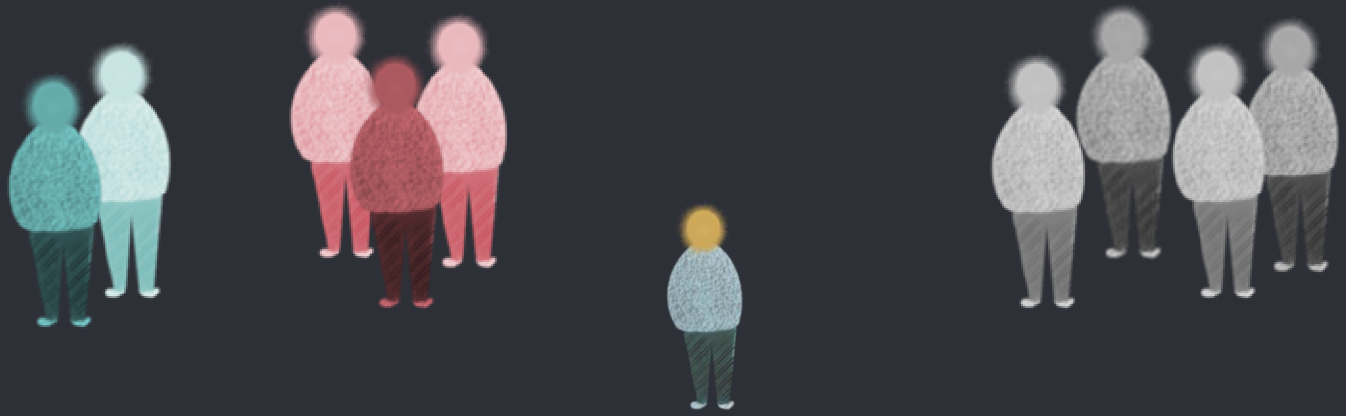
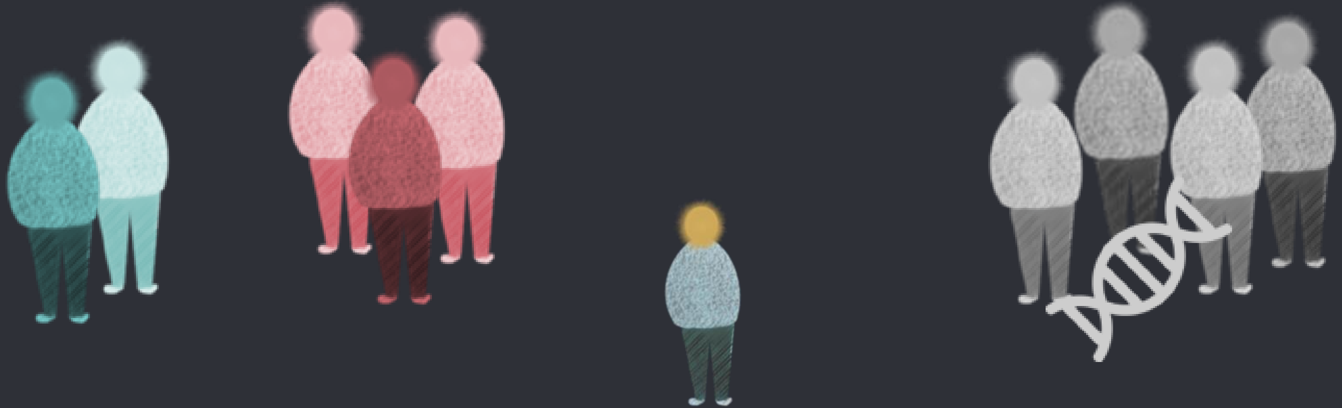
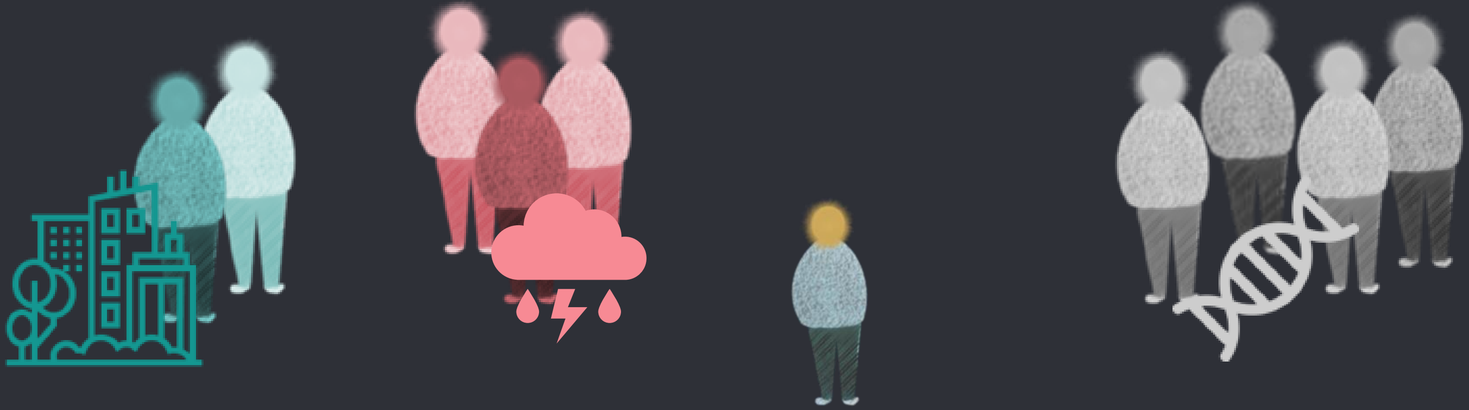
BRAIN CHARTING FOR PRECISION PSYCHIATRY



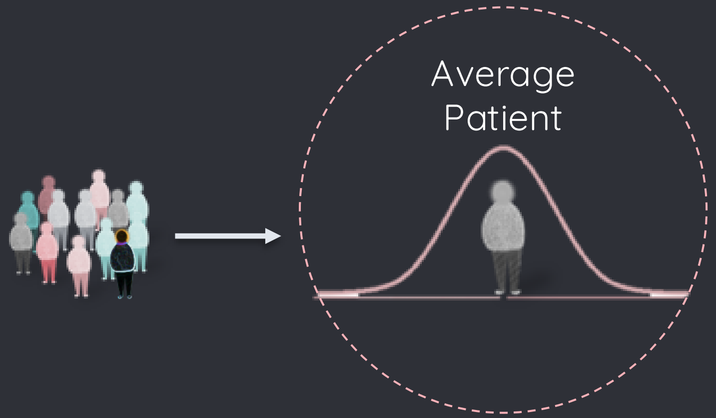




STANDARDIZED MEDICINE

Patient Group

STANDARDIZED MEDICINE



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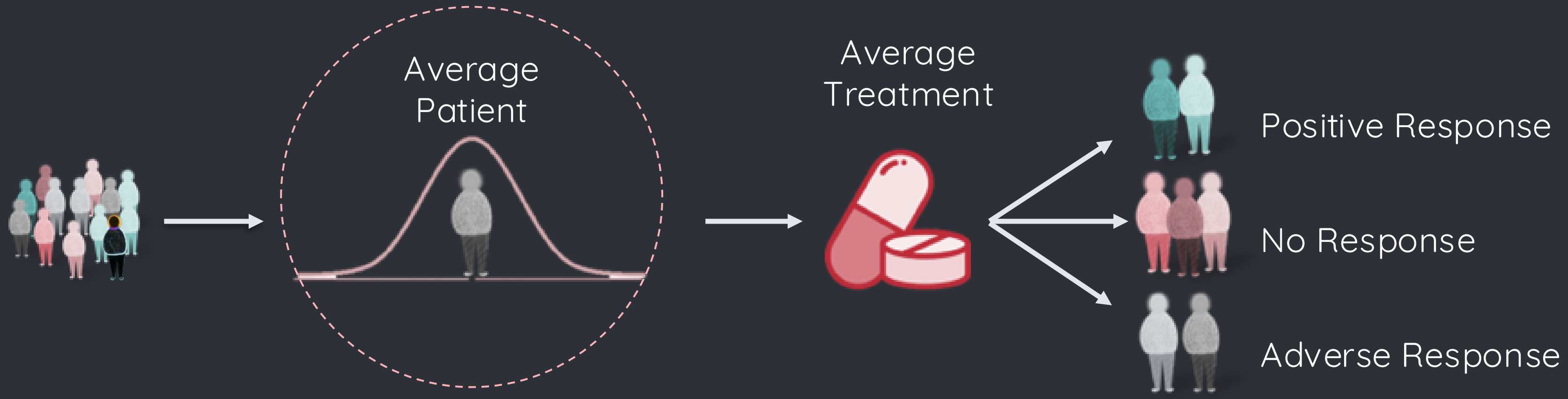
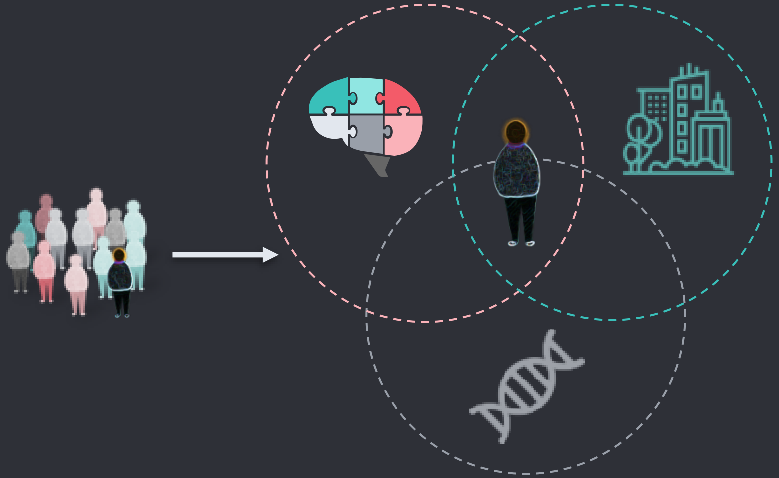


PRECISION MEDICINE



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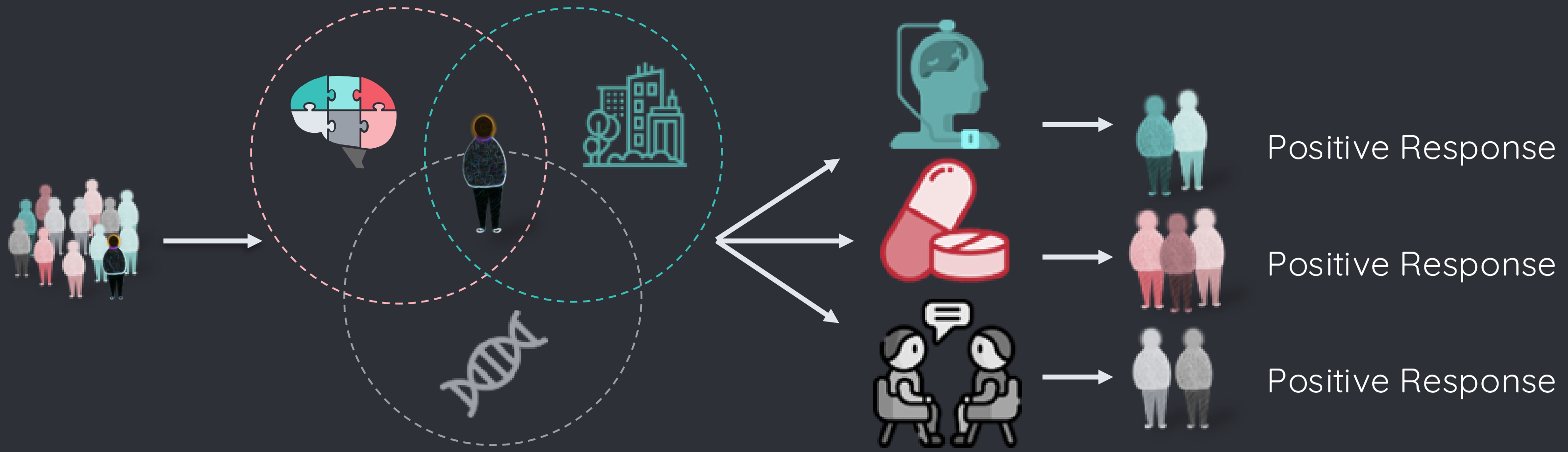
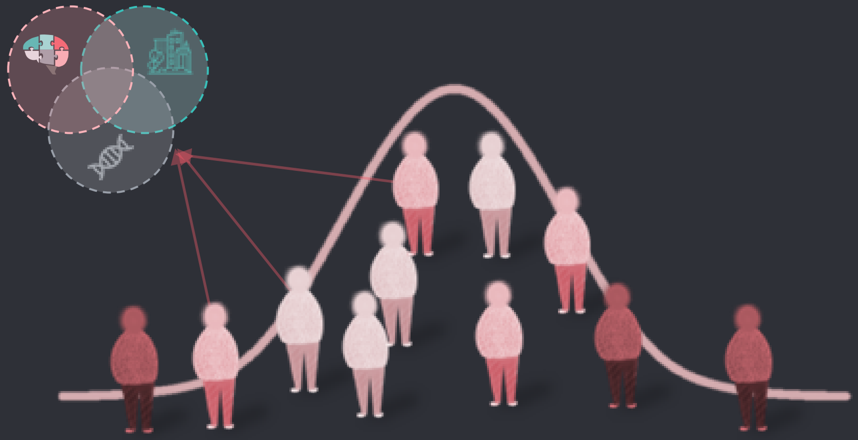


BRAIN CHARTING

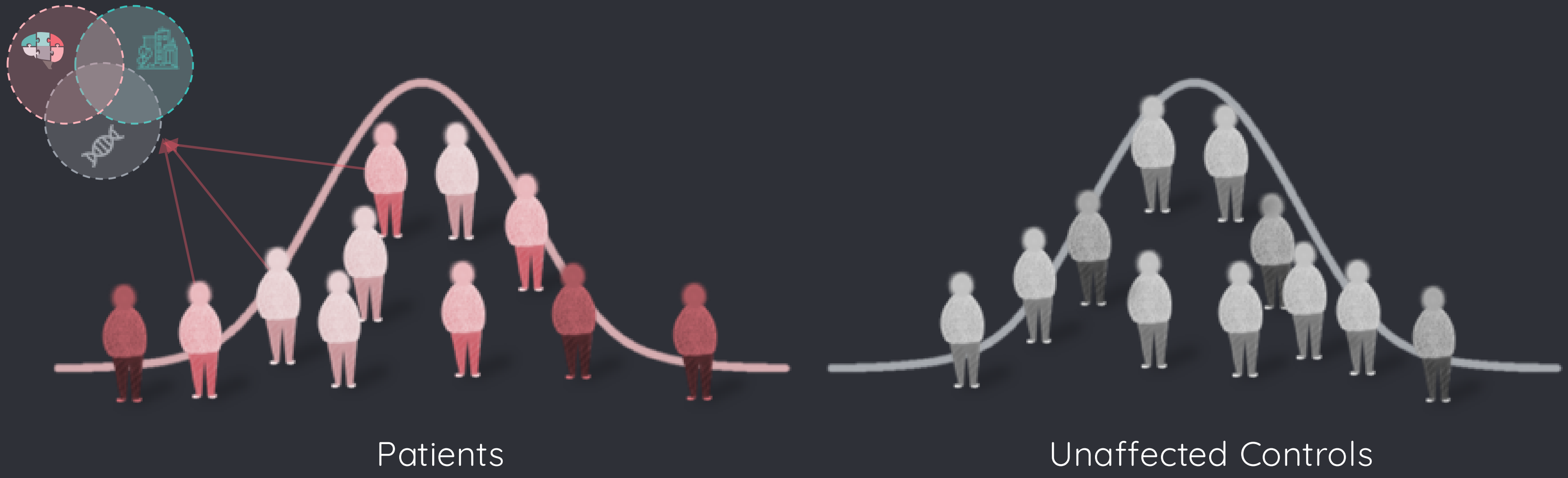
Through Normative Modeling

CASE-CONTROL ASSUMPTION



Patients

CASE-CONTROL ASSUMPTION



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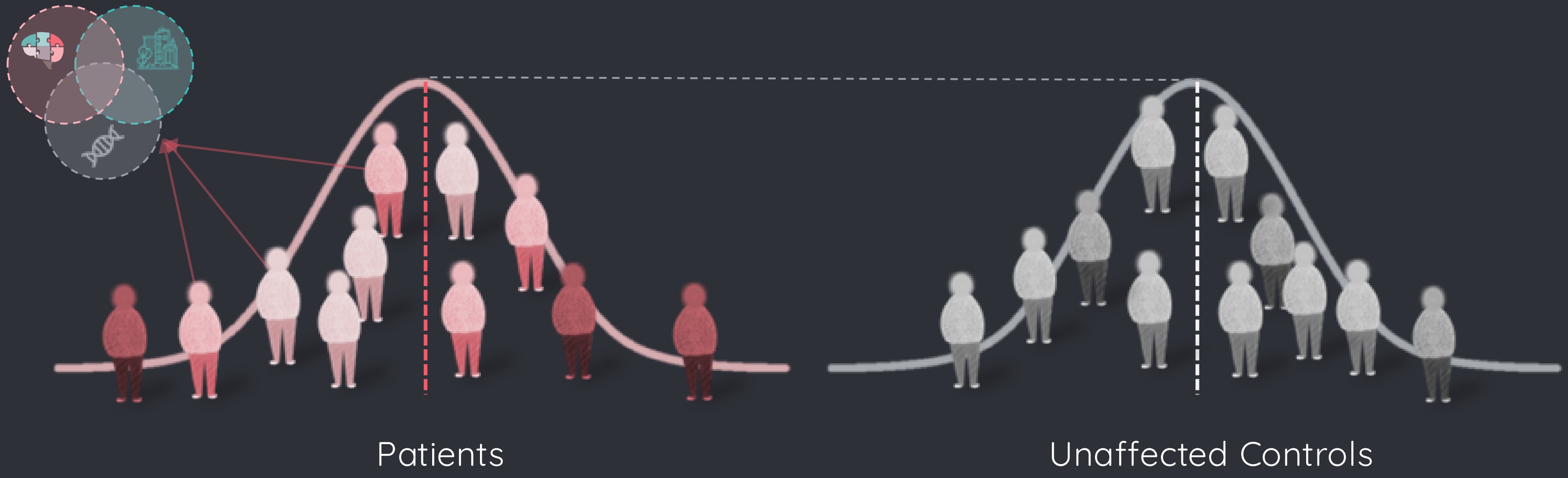
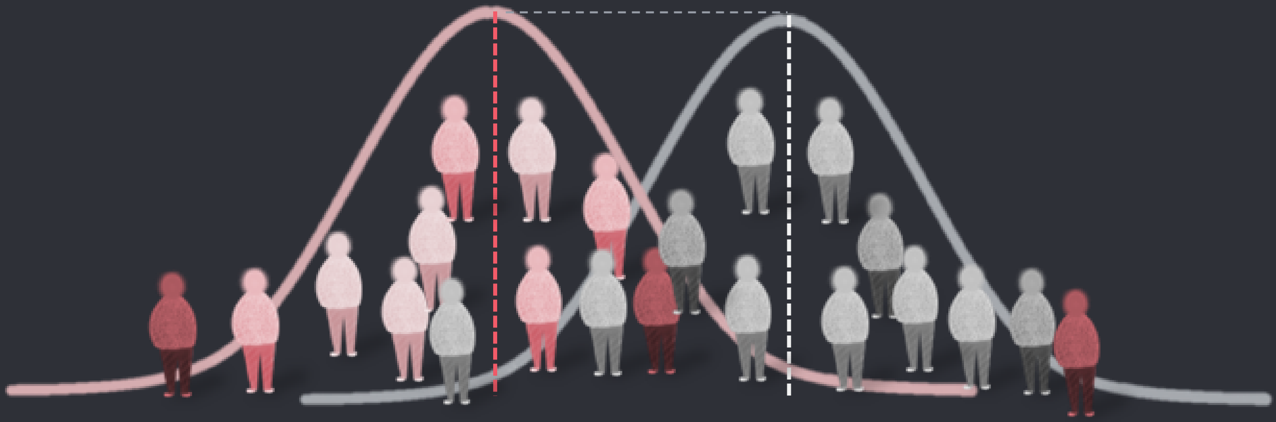


HETEROGENEITY ASSUMPTION



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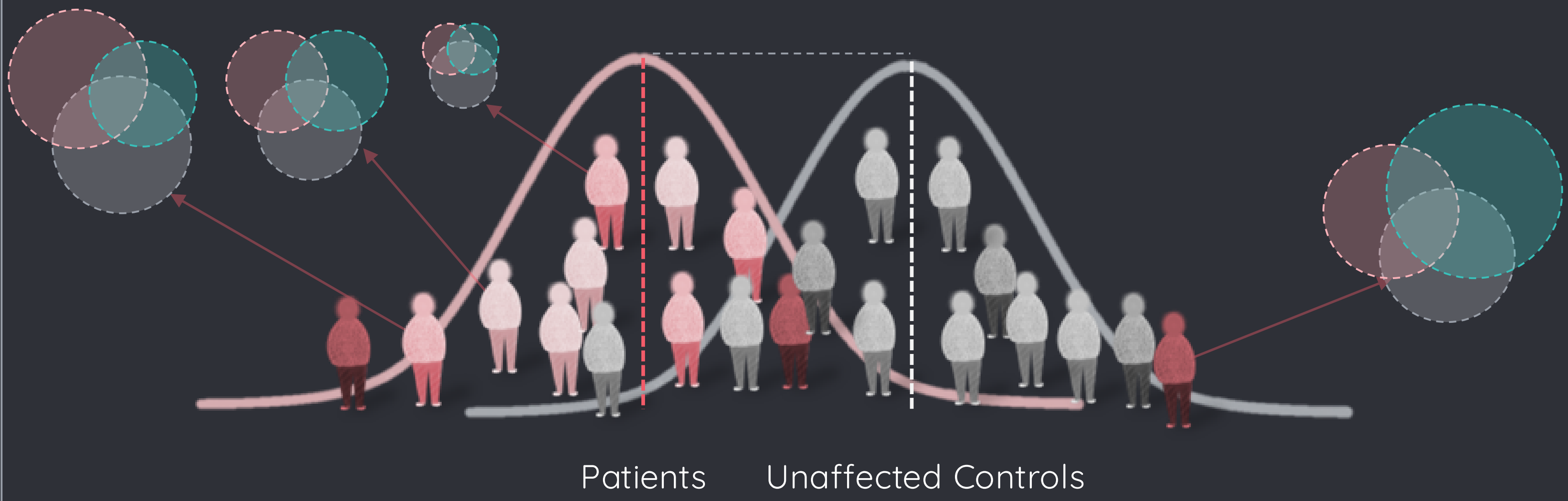

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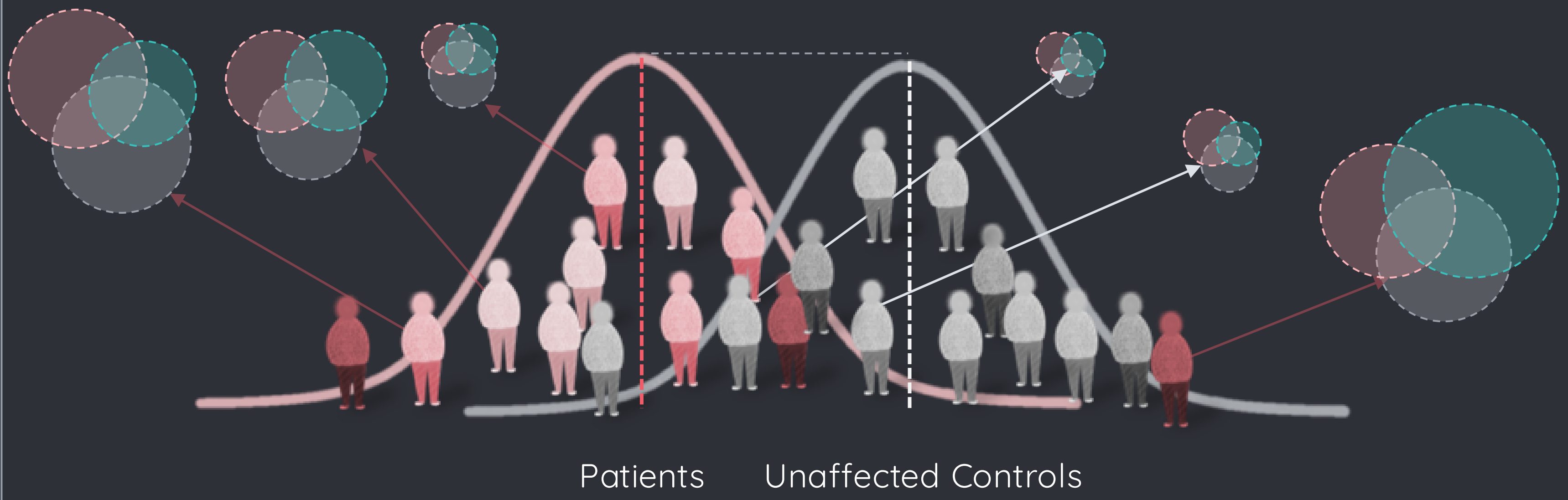
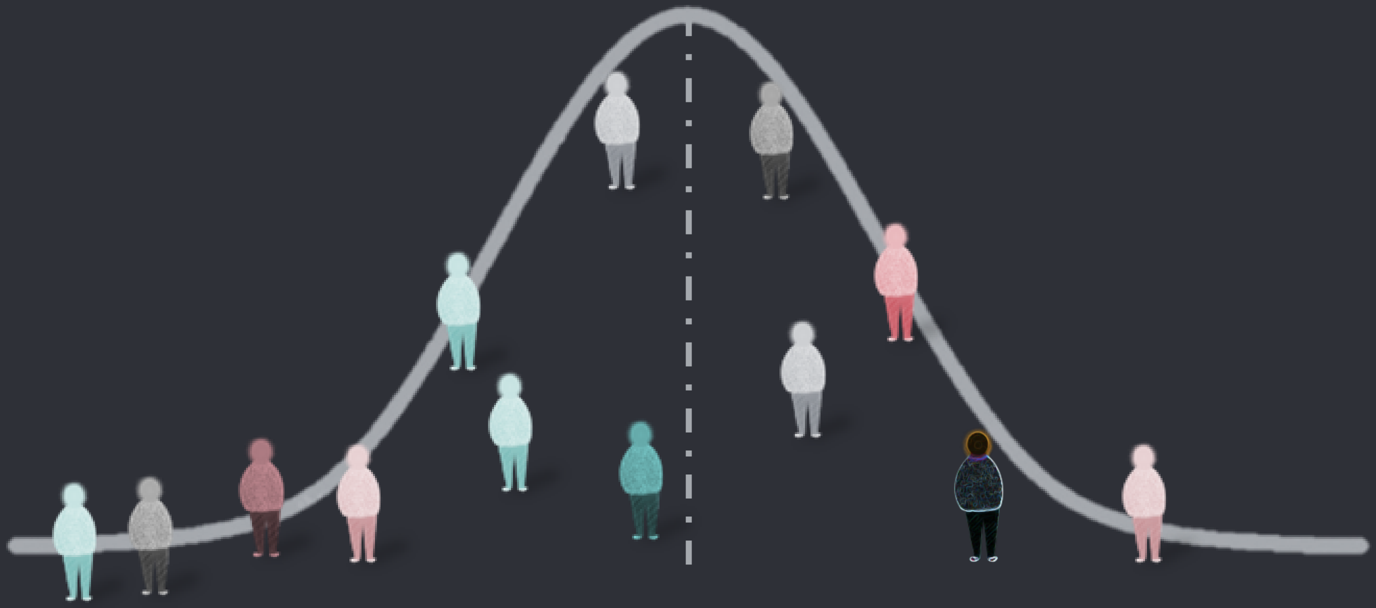


SOLUTION: NORMATIVE MODELS



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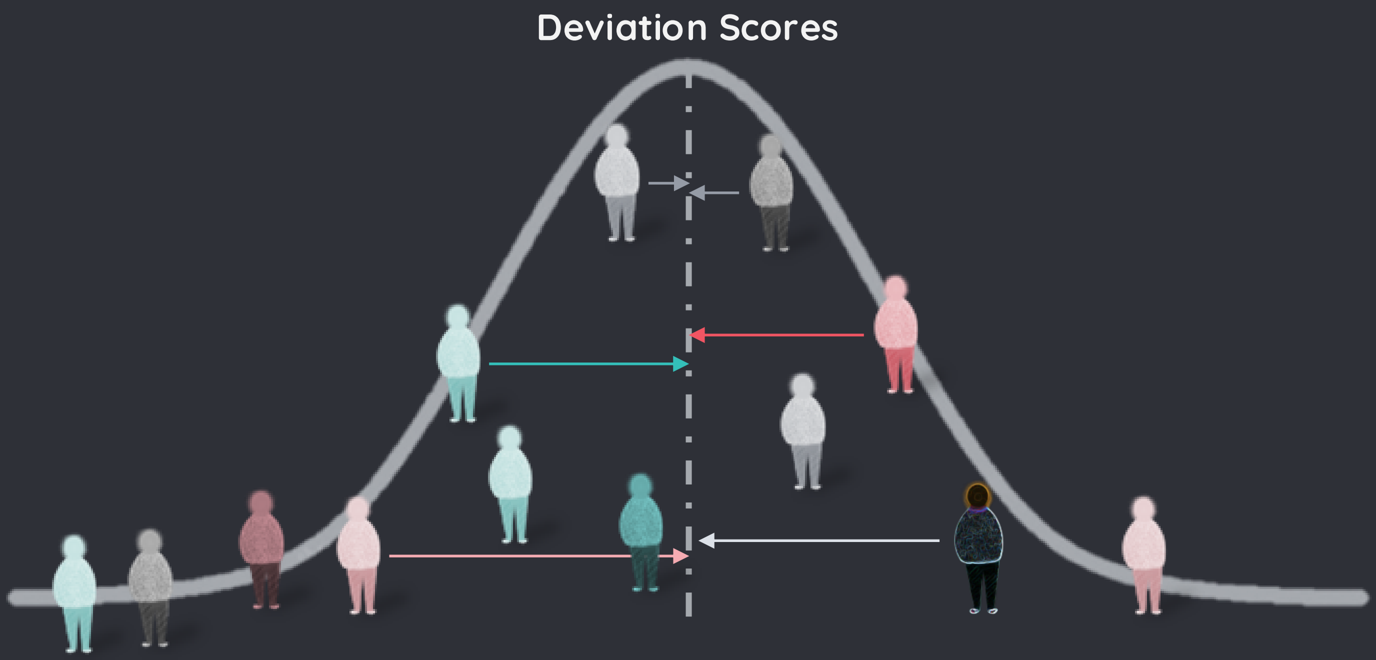

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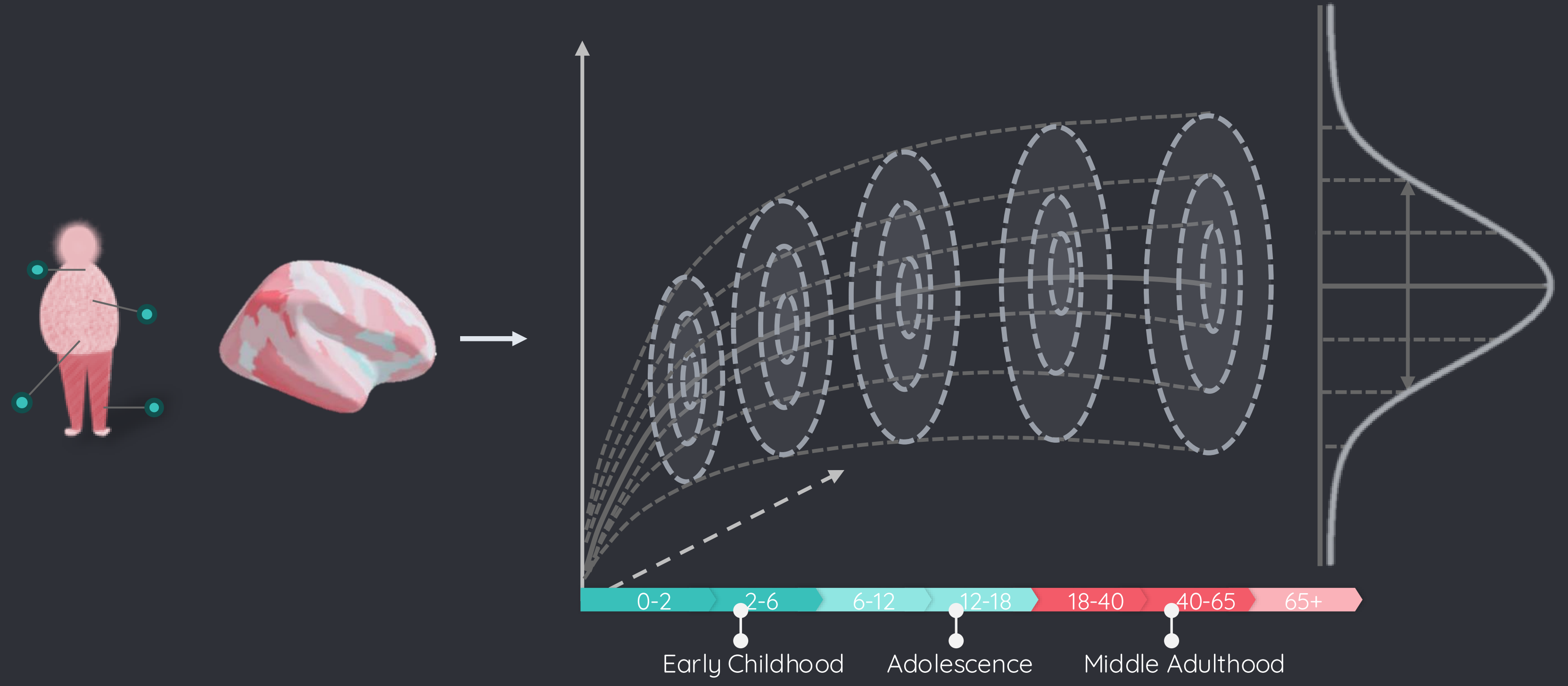

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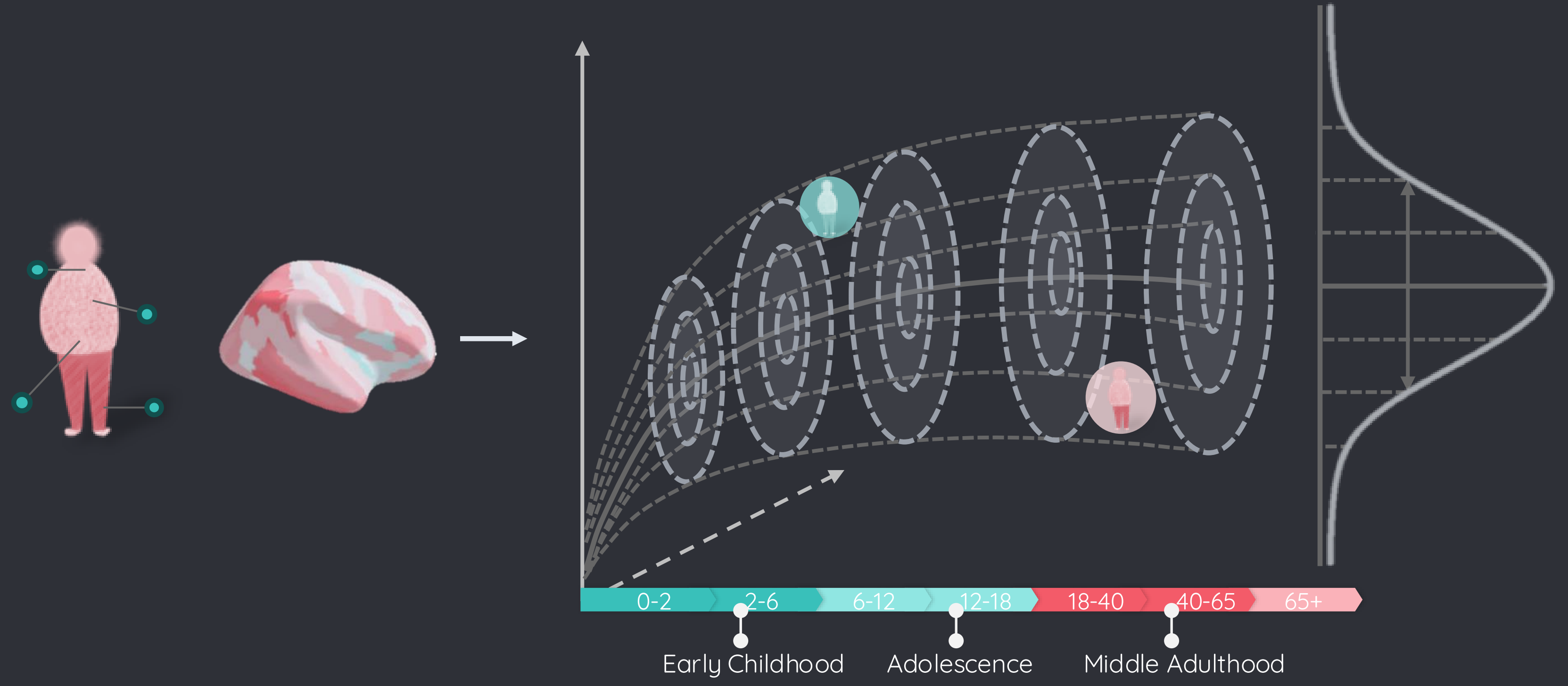
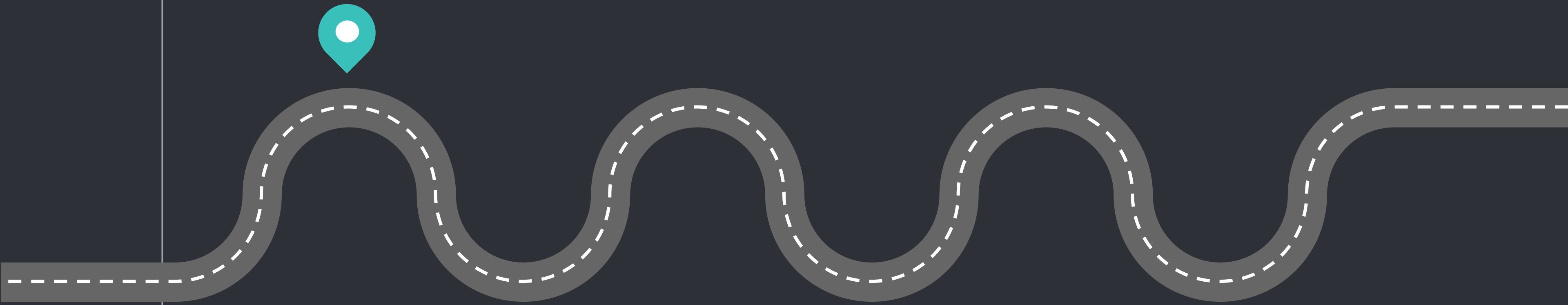


Chapter 1: Why Precision Medicine and Normative Models?



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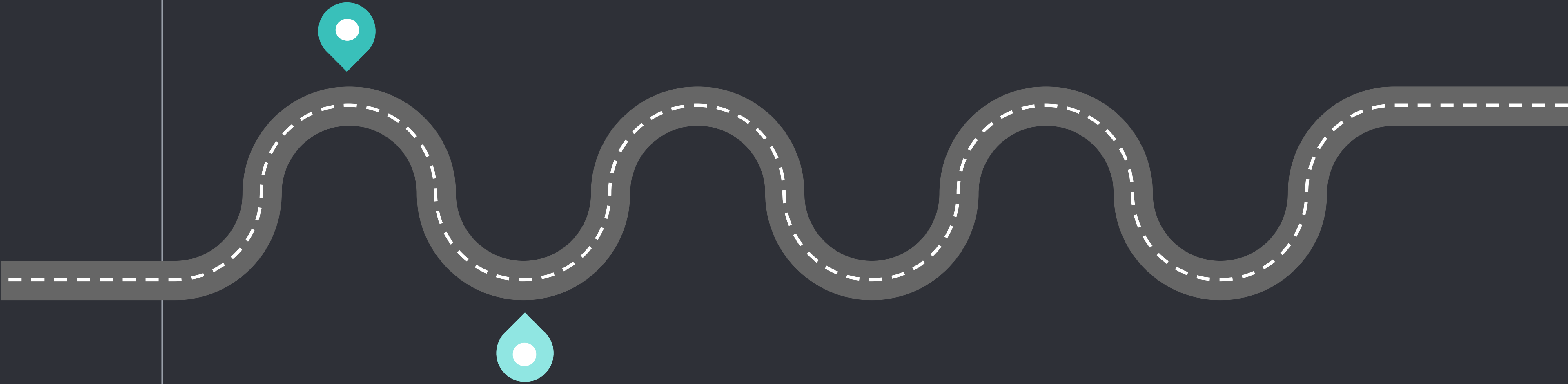
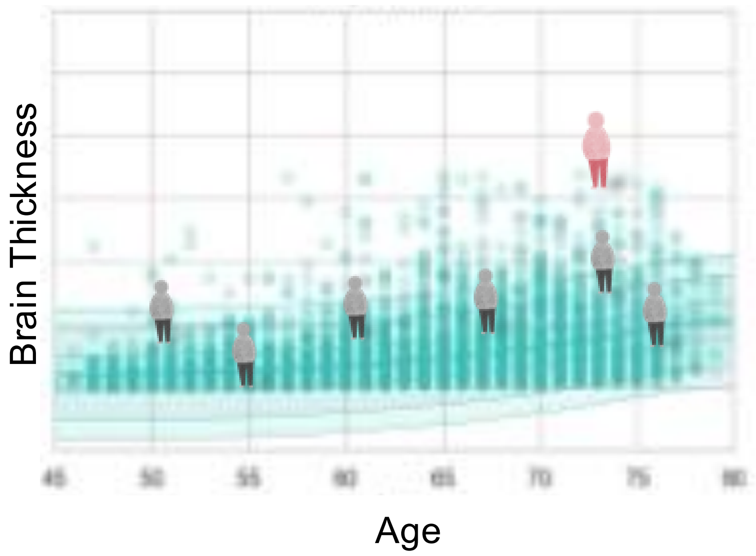


Chapter 2:
Warped
Normative
Models

Chapter 2: Warped Normative Models

Standard Shape Normative Model



**Warped bayesian linear regression for normative modelling of big data. Neuroimage. Fraza et al. 2022*

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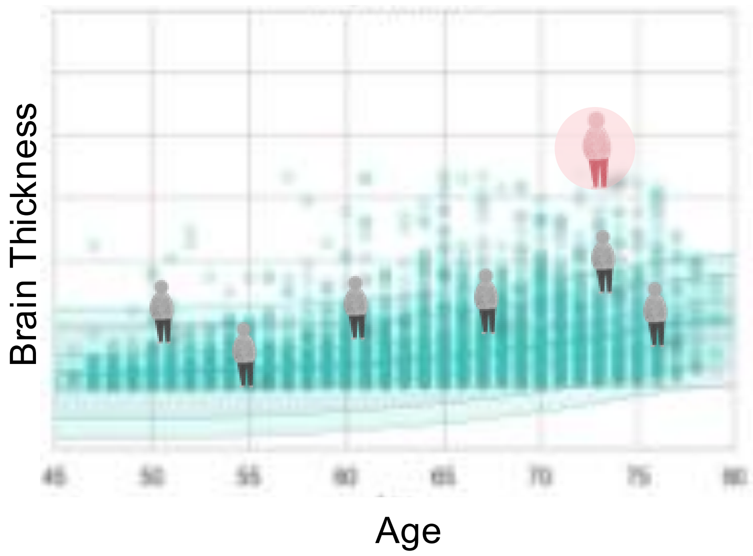

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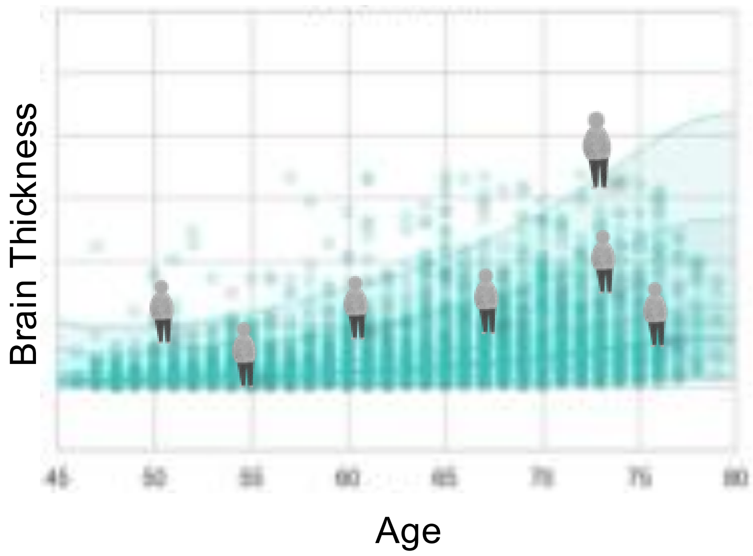
Warped Normative Model



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Chapter 2:
Warped
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Models

Chapter 1:
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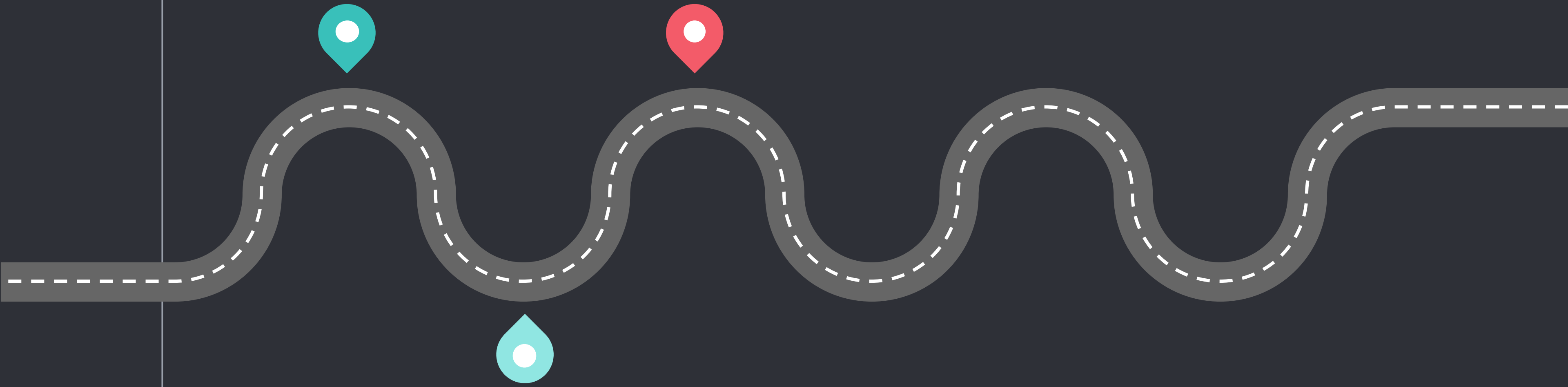
Chapter 3:
Extremes of
Normative Models



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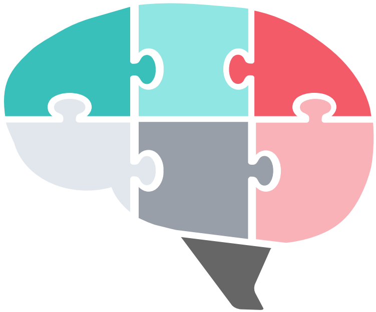
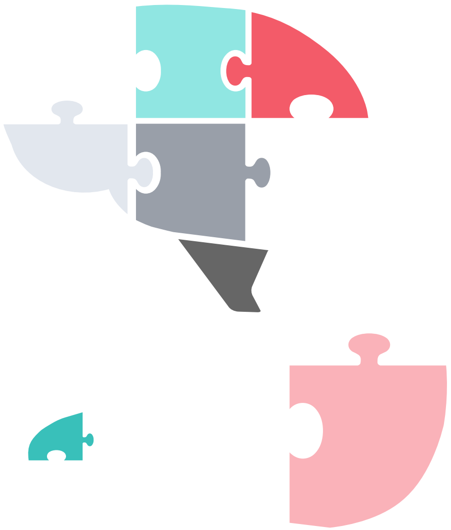
Extreme Values



**Using extreme value statistics to reconceptualize psychopathology as extreme deviations from a normative reference model. Human brain mapping. Fraza et al. 2025.*

Chapter 3: Extremes of Normative Models

Extreme Values



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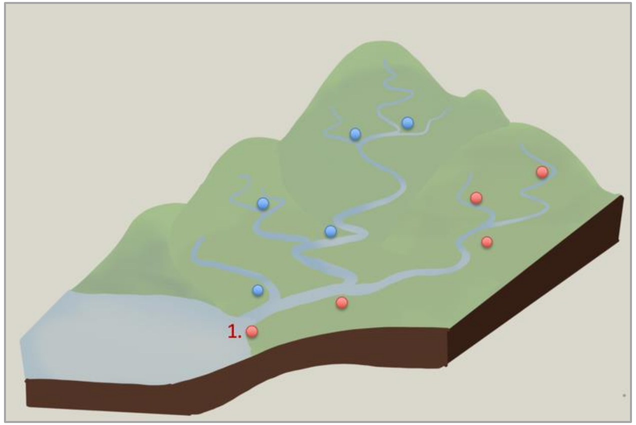
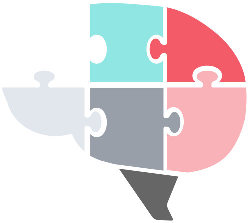
Chapter 3: Extremes of Normative Models

Flood Risks





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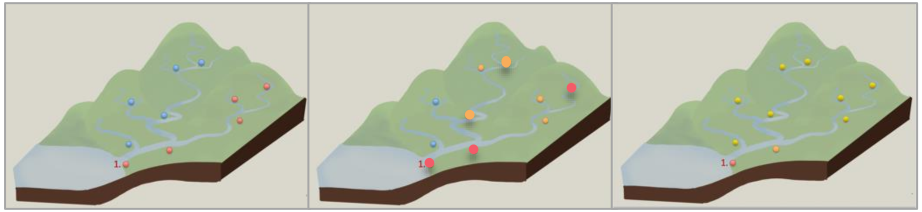
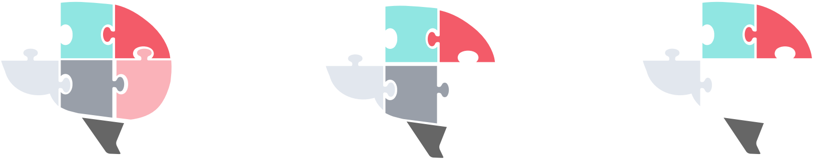
Multiple Extremes





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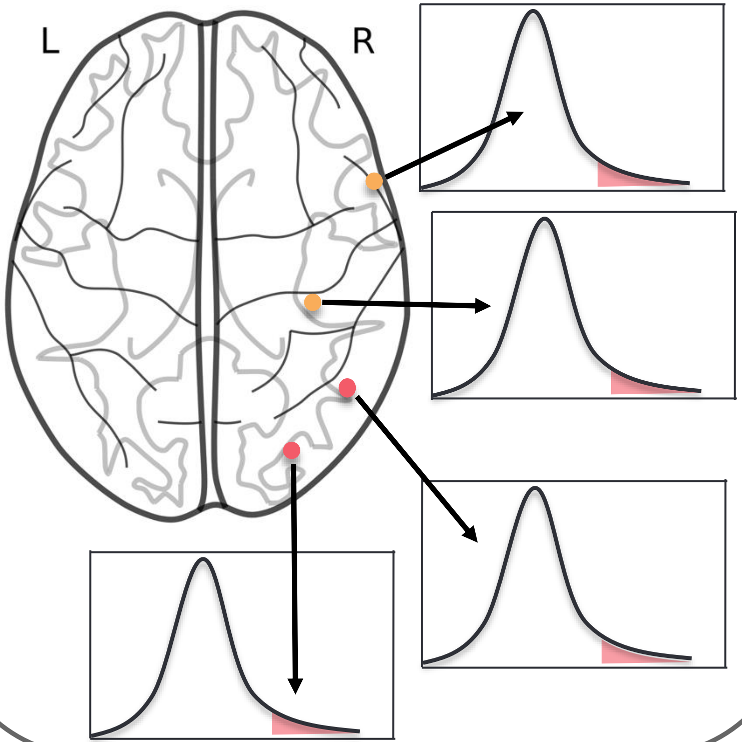

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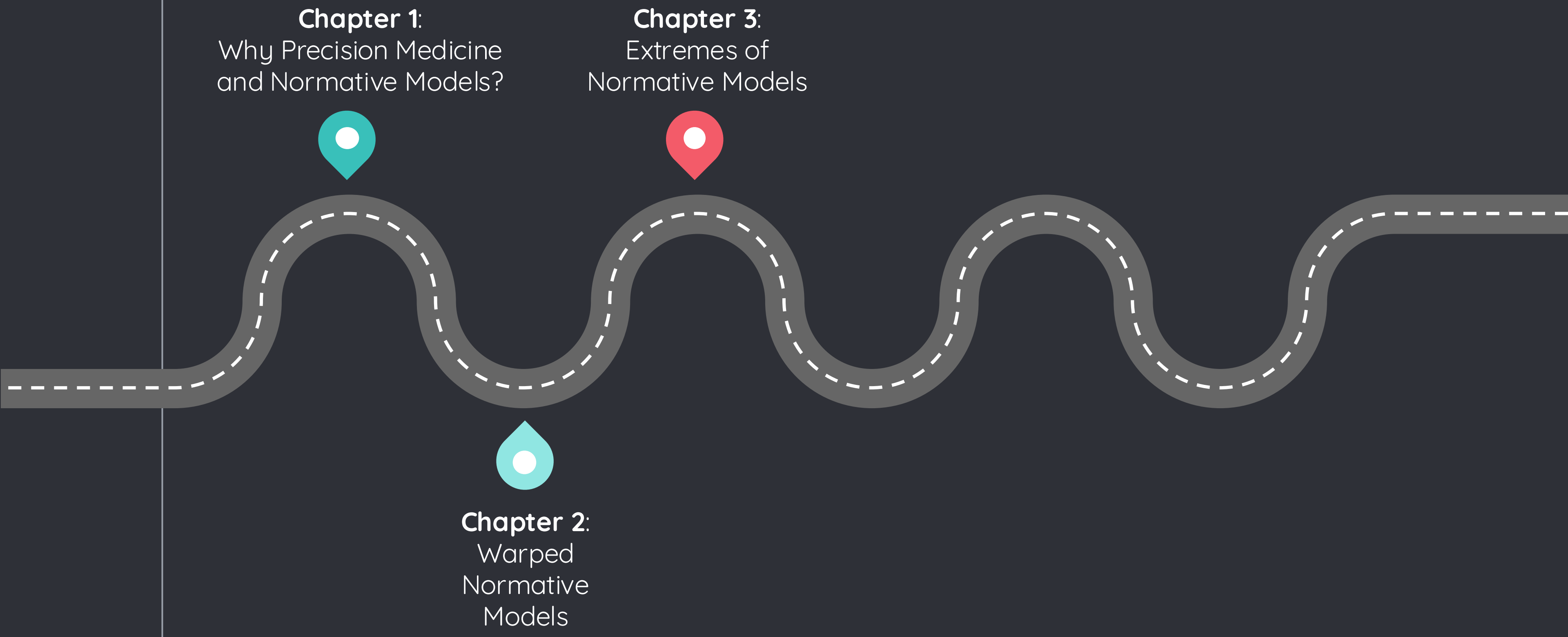
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Chapter 4:
Genetic Application Copy Number Variants

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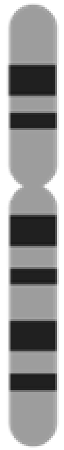
DNA



**Unravelling the link between CNVs, cognition and individual neuroimaging deviation scores from a population-based reference cohort. Nature Mental Health. Fraza et al. 2024.*

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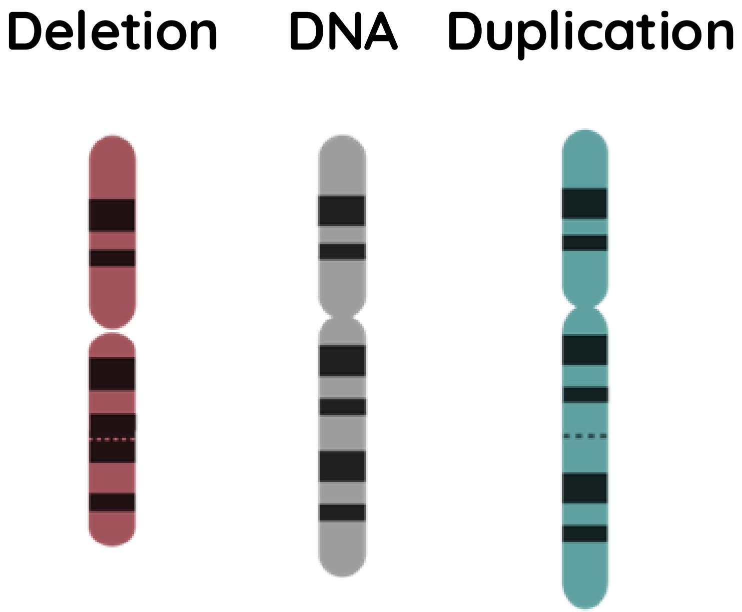

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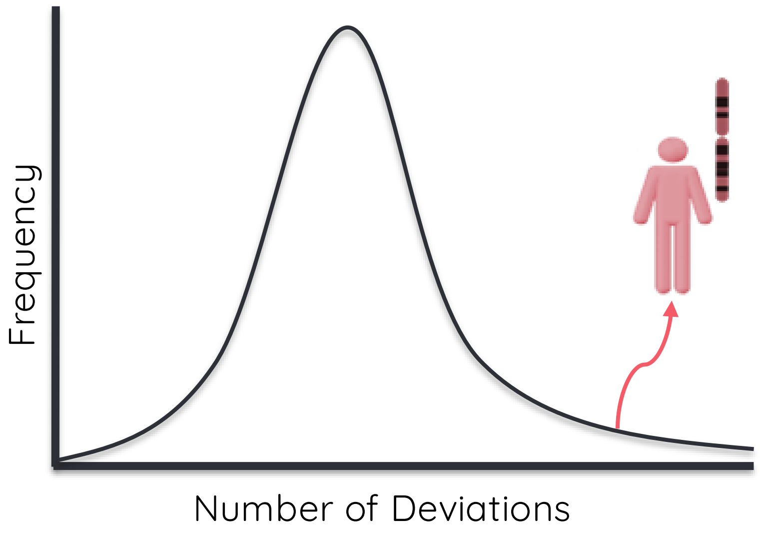
Chapter 4: Genetic Application Copy Number Variants

Compare Individual with CNV to the Population



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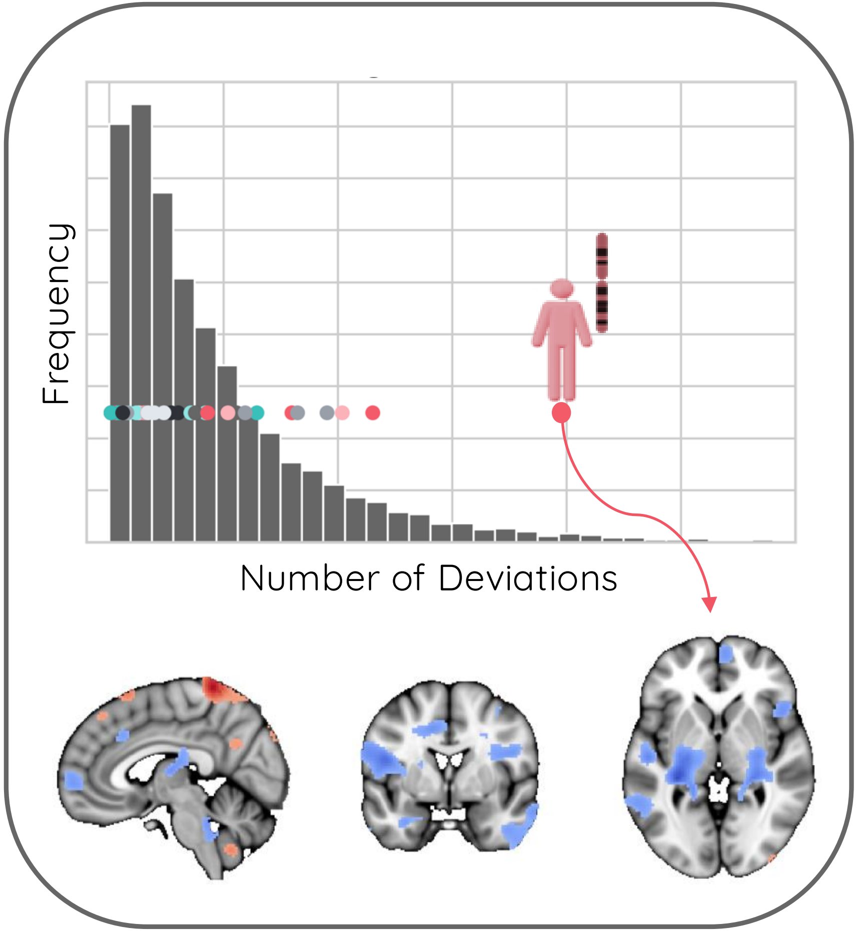

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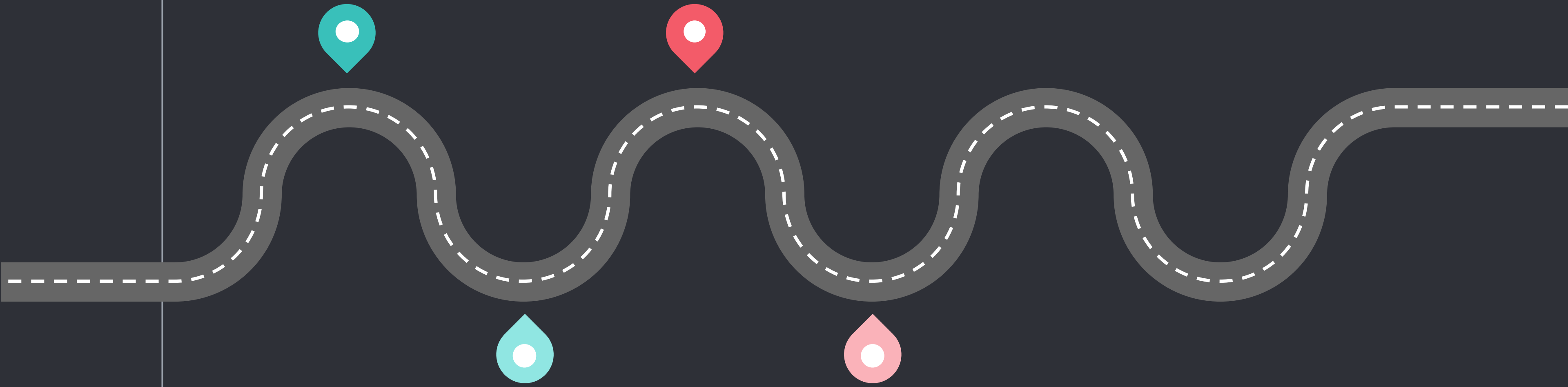

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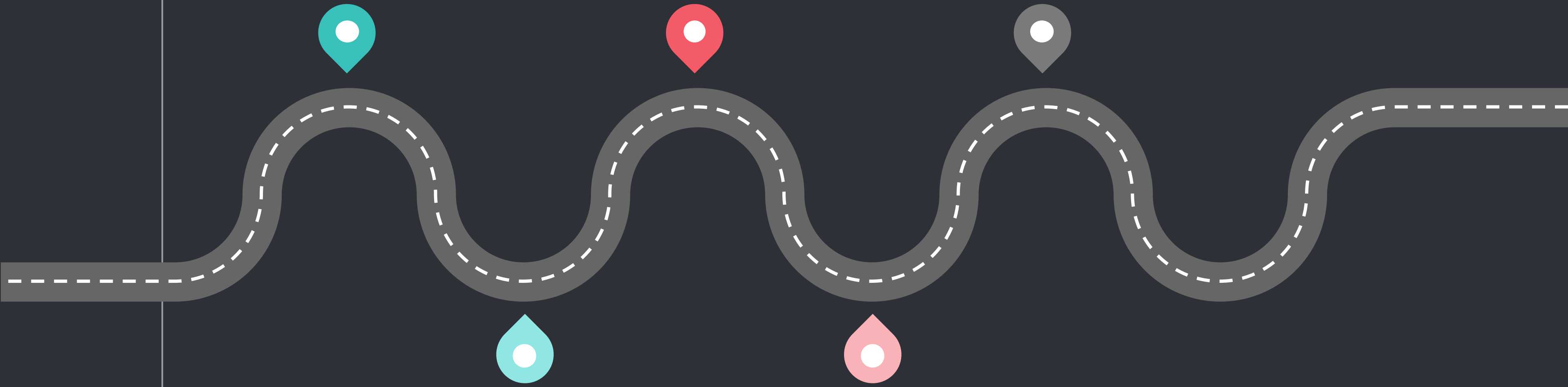
Chapter 5:
Clinical Application Parkinson’s Disease



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Parkinson's Disease

**Individual neurodegenerative progression in Parkinson's disease: A normative modelling Approach. Scientific Reports. Fraza et al. 2025.*

Chapter 5:
Clinical Application
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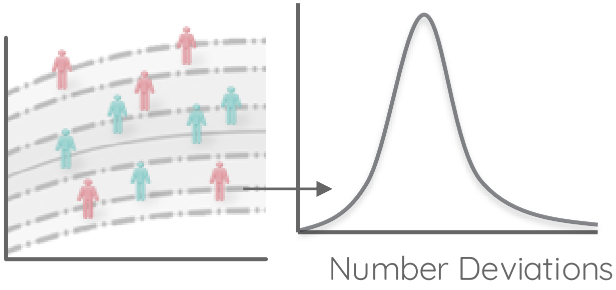
Parkinson's Disease Individual Profile



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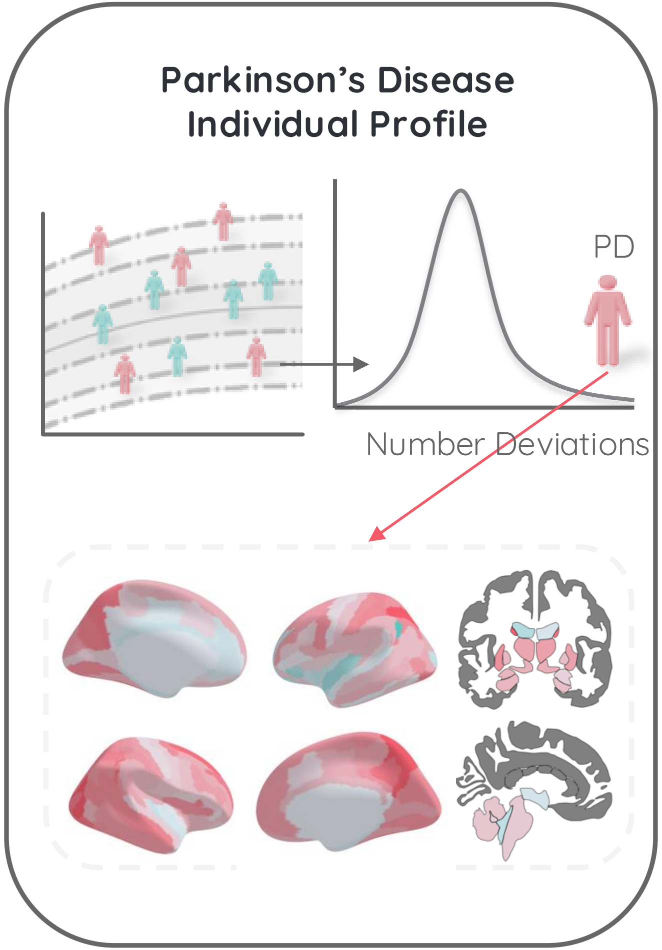

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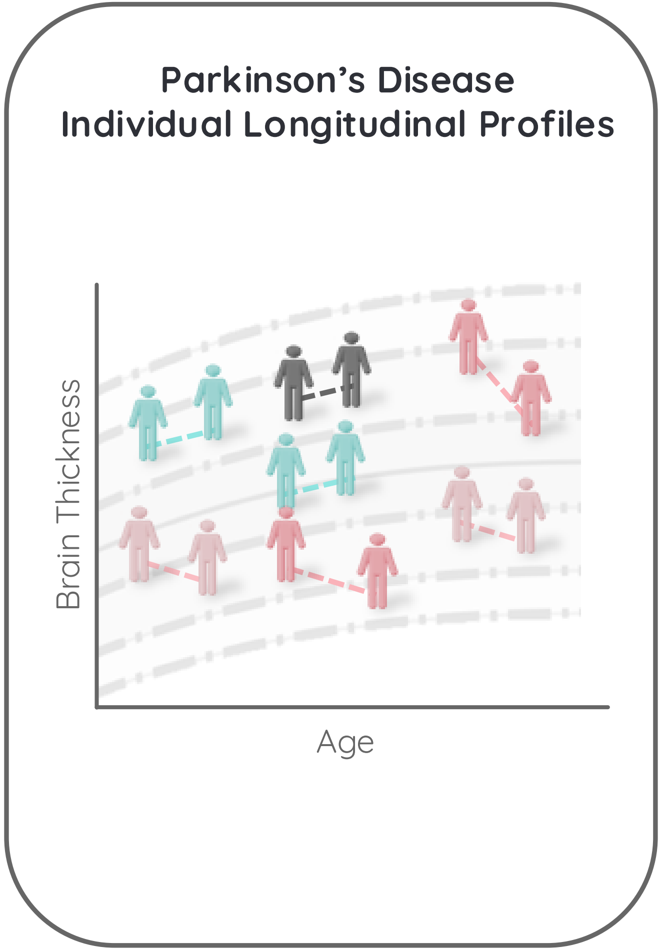

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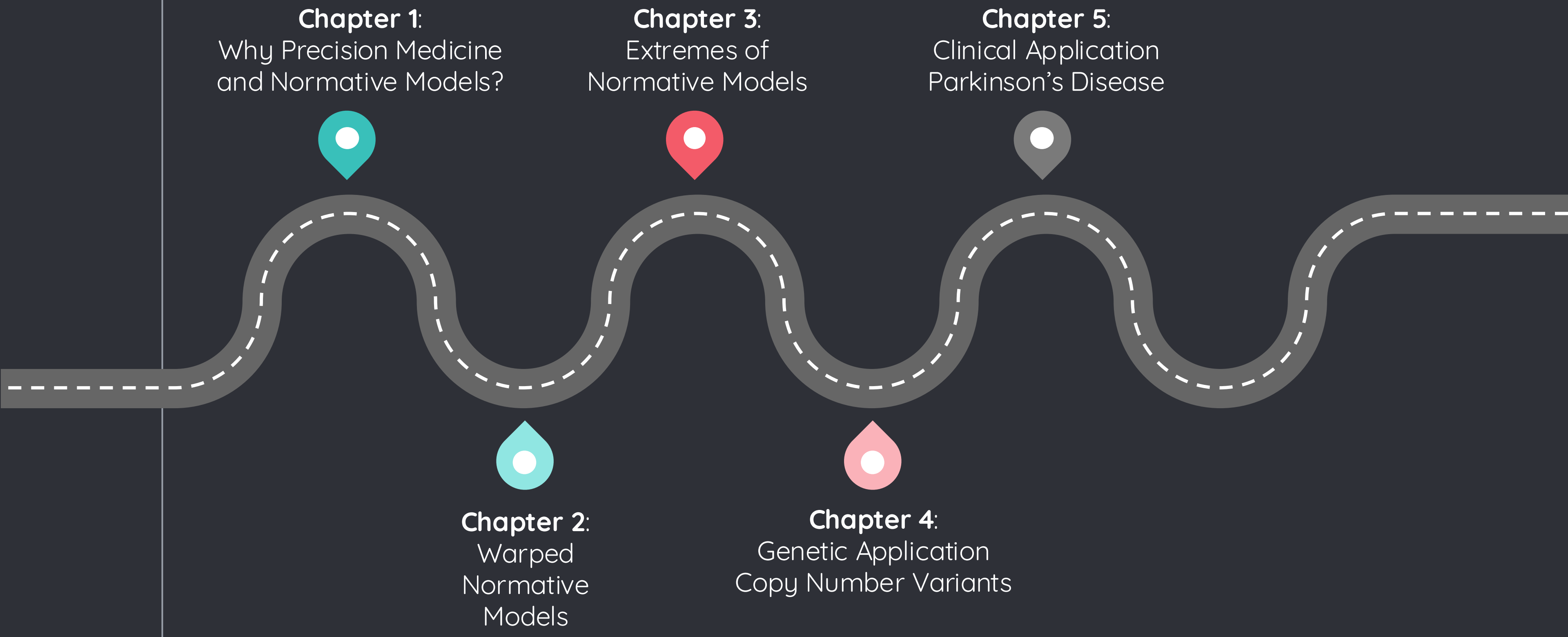
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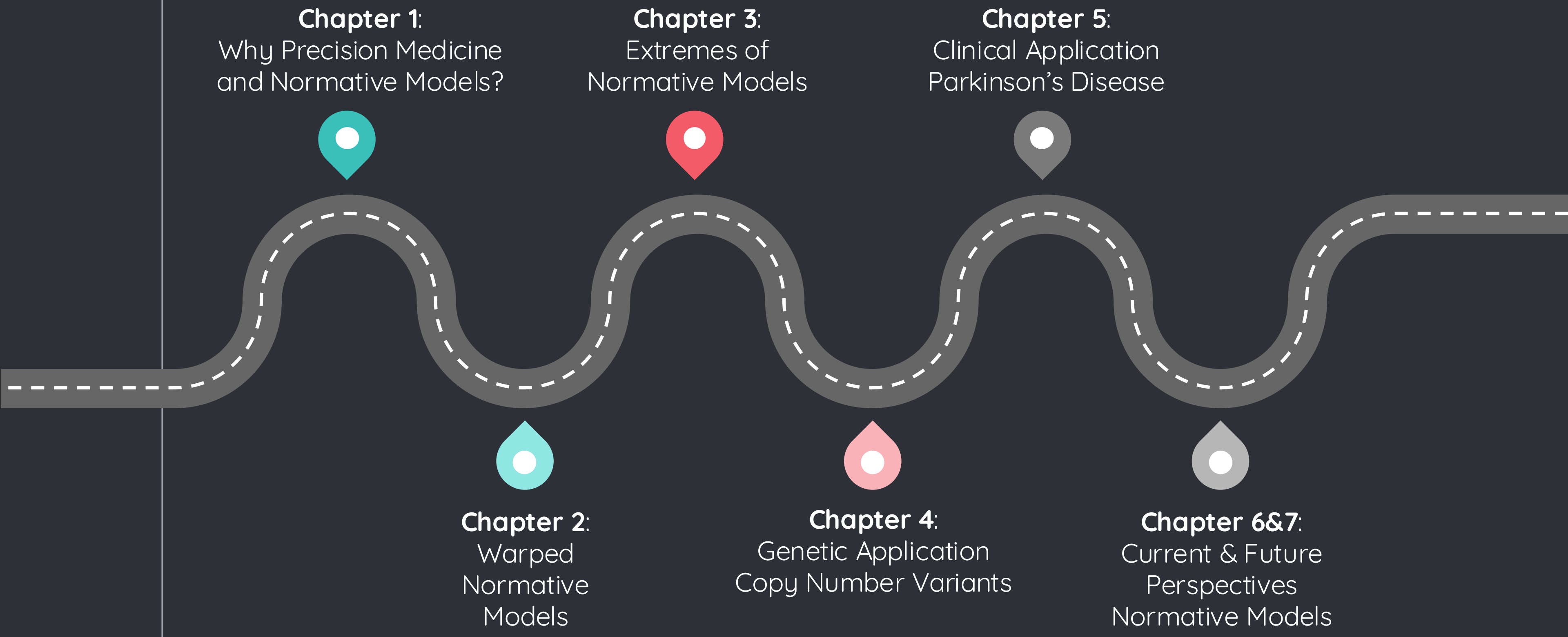
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Chapter 6&7:
Current & Future
Perspectives
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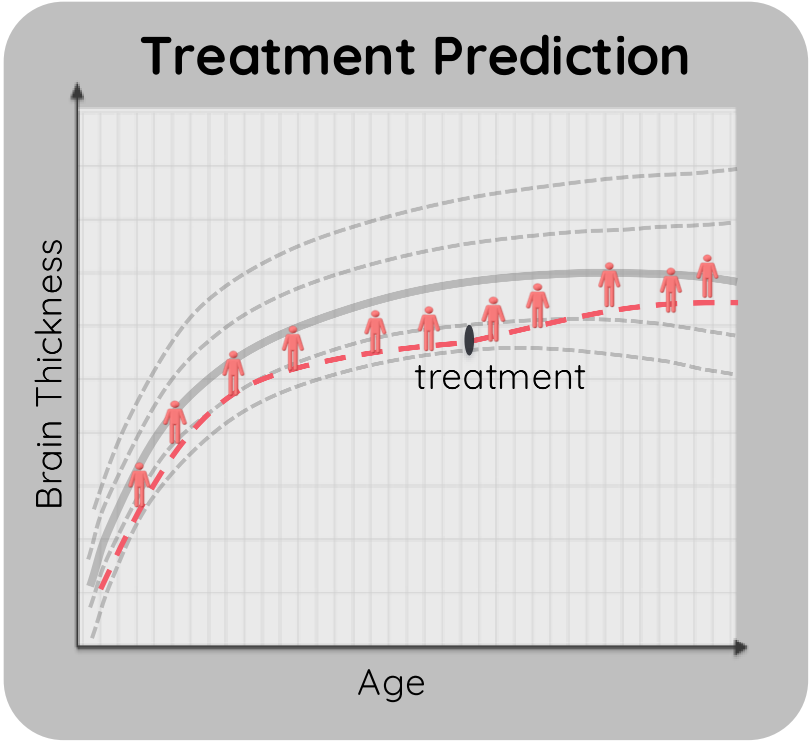
Chapter 6&7: Current & Future Perspectives Normative Models



**The promise of quantifying individual risk for brain disorders through normative modeling, a narrative review. Neurosci Biobehav Rev. Fraza et al. 2025.*

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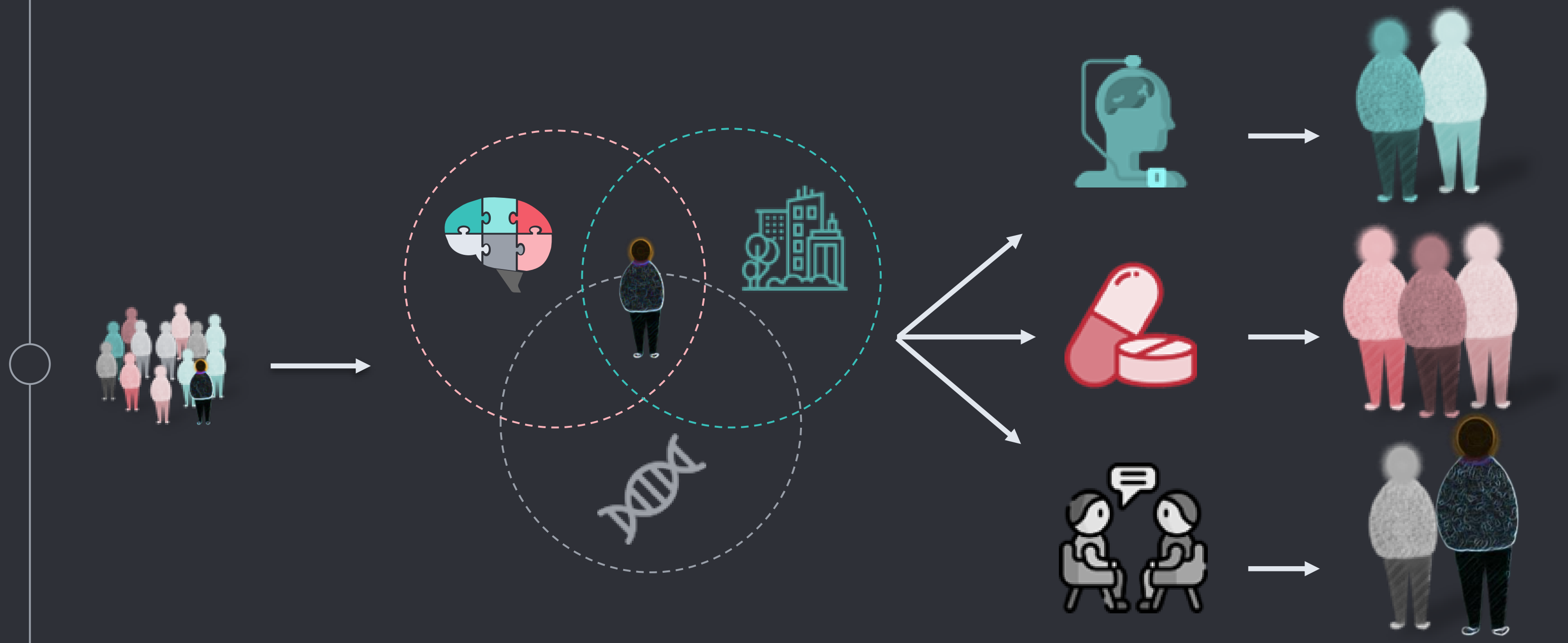

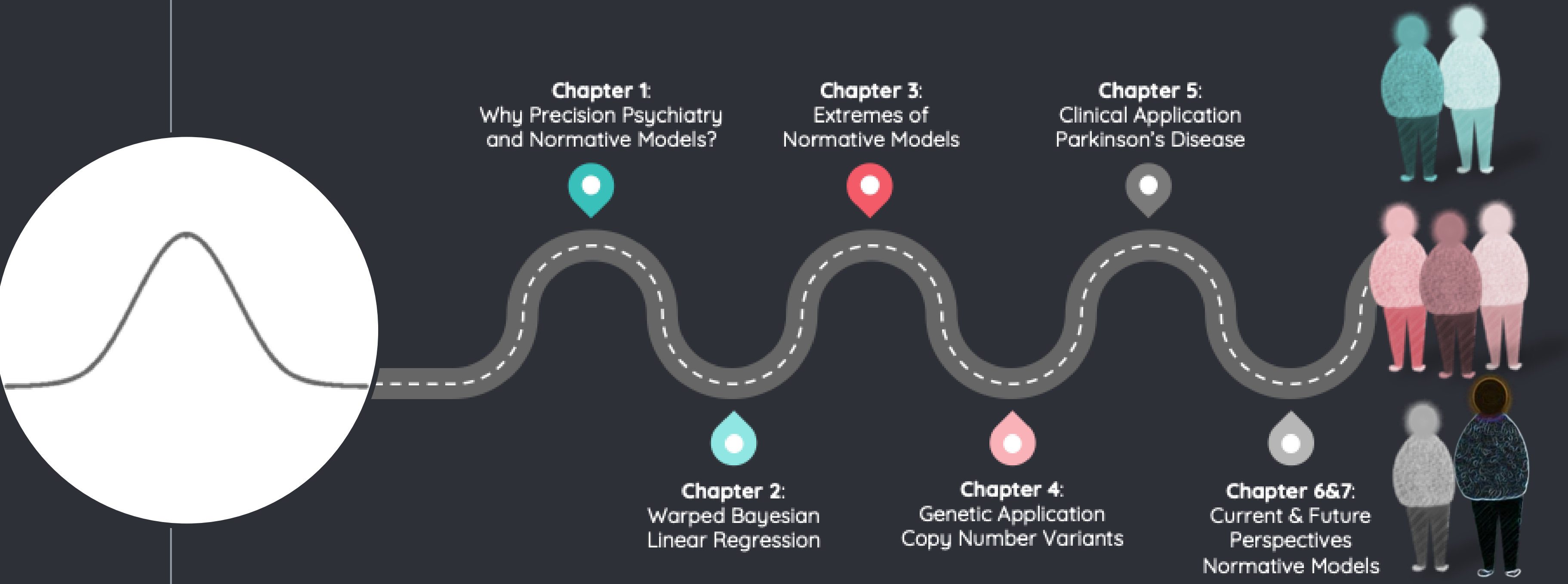

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